Puli Dinesh

ai23btech11019@iith.ac.in | dineshp1601@gmail.com | +91 8463961474 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.89
IGNITE JUNIOR COLLEGE	Class XII (TSBIE)	2022	97.20%
RAVINDRA BHARATHI SCHOOL	Class X (BOARD OF SECONDARY EDUCATION TELANGANA)	2020	10.00%

WORK EXPERIENCE

Deccan AI -- AI Training & Evaluation Freelancer:

- Worked across **4 AI training projects**, evaluating LLM outputs and reasoning traces across Python, SQL, coding, and problem-solving tasks through structured annotation and quality checks.
- Analyzed **1,000+ LLM outputs**, identifying logical, reasoning, and implementation errors, correcting faulty solutions, and refining responses against expected requirements.
- **Benchmarked multi-agent LLM systems with up to 5 agents**, comparing code correctness, reasoning quality, and task-specific performance to provide structured feedback for iterative model training.

PROJECTS

Retrieval-Augmented Document Intelligence System|*Python, Gemini API, FAISS, Sentence Transformers, Gradio*

- Engineered a **RAG pipeline** for PDF, TXT, and CSV documents, combining semantic embeddings with FAISS-based vector retrieval for context-aware question answering.
- Integrated **Sentence Transformers and Gemini LLMs** to retrieve relevant document chunks and generate context-grounded responses with source attribution.
- Developed an interactive **Gradio interface** supporting multi-document ingestion, semantic search, and natural-language querying.

Deep Reinforcement Learning: DQN & Policy Gradient | *Python, PyTorch, OpenAI Gym:*

- Explored **value-based control with DQN** on MountainCar-v0, learning momentum-driven navigation through experience replay and Q-value optimization.
- Investigated **policy-gradient optimization with REINFORCE** on CartPole-v1 and LunarLander-v2, evaluating Reward-to-Go and Advantage Normalization for variance reduction.
- Analyzed **convergence dynamics and training stability** across batch sizes and variance-reduction strategies, identifying their impact on policy learning performance.

Cultural-Specific Named Entity Recognition | *Python, BERT, Hugging Face*

- Built a 12,000-sample cultural NER dataset using **template-based augmentation** to generate diverse, meaningful samples and BIO-tagged **15+ Indian cultural entity categories** spanning food, festivals, rituals, and traditions.
- Fine-tuned **BERT for token classification**, achieving an **F1-score of 0.8649** on unseen test data while recognizing multiple cultural entity categories.
- Improved contextual entity recognition through **BIO sequence labeling** and balanced data augmentation, enabling detection of **multi-word cultural entities** and underrepresented Indian cultural concepts

Kidney Stone Detection using YOLOv8 | *Python, YOLOv8, OpenCV, Flask*

- Formulated an **object-detection pipeline** with YOLOv8 to localize kidney stones across **1,300+ annotated medical images**, covering training, validation, and test splits.
- Established an **automated computer-vision inference workflow** combining YOLOv8 and OpenCV for image preprocessing, stone localization, bounding-box generation, and result visualization.
- Operationalized model inference through a **Flask-based web interface**, enabling image uploads, real-time predictions, annotated outputs, and interactive detection results.

SKILLS

Programming: Python, C++, C, SQL

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, Flask, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, Google Colab, Jupyter, VS Code, LaTeX

Technologies: REST APIs, Pixhawk, ArduPilot, Fusion 360, Solid Edge, 3D Printing

RELEVANT COURSES

Core Courses: Computer Vision, Natural Language Processing, Deep Learning, Reinforcement Learning, Foundations of Machine Learning, Convex Optimization,Data Science Analysis, Robotics.

Computer Science: Introduction to Programming, Discrete Mathematics, Data Structures, Computer Architecture, Operating Systems, Database Management Systems, Compilers

Mathematics: Numerical Analysis, Introduction to Bayesian Statistics, Calculus, Differential Equations, Series of Functions, Introduction to Metric Spaces, Matrix Theory, Applied Statistics, Probability and Random Variables

SCHOLASTIC ACHEIVEMENTS

Qualified for the Final Round of **HMEL Energy Quest 3.0**, an energy-focused national competition organized by HPCL-Mittal Energy Limited (HMEL).

Participated in the Brand & Buzzwrd Hackathon on Kaggle, competing in a machine-learning challenge focused on generalization to unseen brand names.

Participated in the Kaggle Competition - DL F2025, working on a **semantic segmentation** challenge.

POSITIONS OF RESPONSIBILITY

Sponsorship Coordinator, Milan 2025, IIT Hyderabad

- Coordinated sponsor outreach and communication to support event funding and partnerships.
- Managed **sponsorship coordination and stakeholder engagement** for event execution.

Sponsorship Coordinator , Abhyant 2026, IIT Hyderabad

- Driving **sponsorship outreach and partner engagement** for Abhyant 2026.
- Facilitating **sponsorship coordination and external stakeholder communication** as part of the organizing team.

EXTRACIRRICULAR

Strategic & Competitive Pursuits: Engage in **Chess, Badminton, Table Tennis, Free Fire, 8 Ball Pool, and PC/Mobile Gaming**, combining strategic thinking with competitive gameplay.

Gold Medalist - Free Fire (Team Event): Represented the hostel team at **Milan Inter-Hostel Competitions**, securing **1st place** through coordinated team play and competitive strategy.

Creative Arts: Training in **Music, Painting & Poetry** under *Oopali Operajita*.